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A clinically accessible heart disease prediction framework: multi-model evaluation with ensemble learning and low-code deployment

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Abstract

Heart disease remains a leading global cause of mortality, highlighting the urgent need for effective early diagnostic tools. This study introduces a robust, comparative machine learning framework for predicting heart disease based on a consolidated dataset comprising 918 patient records and 46 clinically relevant features. Ten well-established supervised learning algorithms—including Gradient Boosting, Random Forest, Logistic Regression, Support Vector Machine (SVM), Neural Network, AdaBoost, CN2 Rule Induction, k-Nearest Neighbors (kNN), Naive Bayes, and Decision Tree—were rigorously evaluated. The models were assessed using a suite of metrics, including accuracy, precision, recall, F1-score, area under the curve (AUC), and Matthew's correlation coefficient (MCC), to ensure a comprehensive performance profile. Gradient Boosting achieved the highest predictive accuracy (87.4%) and AUC (0.928), outperforming all other models in identifying patterns within the clinical dataset. The methodology integrates both Python-based libraries and the Orange Data Mining tool to support low-code, reproducible workflows for healthcare practitioners and researchers. In addition to delivering high-performance classification, the study highlights model interpretability, feature relevance, and practical deployment using accessible platforms. These contributions underscore the potential of ensemble-based machine learning to enhance early detection and clinical decision-making in cardiovascular healthcare. This study distinguishes itself by integrating a low-code, clinician-friendly deployment pipeline using Orange Data Mining alongside Python, enabling real-time, interpretable risk assessment without requiring deep programming expertise. The framework emphasizes transparent feature relevance and multi-metric evaluation to support clinical decision-making.

Keywords Heart disease prediction, Gradient boosting, Machine learning models, Ensemble methods, Clinical decision support, Cardiovascular diagnosis, Supervised learning, Medical data classification

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1 Introduction

Cardiovascular diseases (CVDs) continue to be the leading cause of mortality worldwide, responsible for an estimated 17.9 million deaths annually, or 32% of all global fatalities, as reported by the World Health Organization [1]. The complexity and often-subtle presentation of heart-related conditions pose significant diagnostic challenges, particularly in early stages where timely intervention can be lifesaving. Consequently, developing intelligent, data-driven diagnostic systems has become a priority in modern healthcare.

Traditional diagnostic frameworks rely on physician expertise, electrocardiographic interpretation, and manual clinical assessments that, while valuable, are often subjective and inefficient [2]. With the rise of electronic health records and accessible computational resources, machine learning (ML) has become a transformative solution in automating disease prediction and enhancing clinical decision support. Ensemble learning techniques have demonstrated strong performance in identifying complex patterns across high-dimensional medical data [3].

Despite progress, many existing studies are constrained by narrow algorithm comparisons or limited datasets, thus lacking generalizability in real-world applications [4]. Moreover, comprehensive performance evaluations of modern ensemble models such as Gradient Boosting remain underexplored in the context of heart disease diagnostics. Gradient Boosting, known for its ability to minimize bias and variance through iterative refinement, presents a promising solution for improving prediction accuracy [5].

To address these gaps, this study conducts a systematic comparative analysis of ten prominent machine learning algorithms—including Gradient Boosting, Random Forest, Logistic Regression, Support Vector Machine (SVM), Neural Network, AdaBoost, CN2 Rule Induction, k-Nearest Neighbors (kNN), Decision Tree, and Naive Bayes—on a consolidated heart failure dataset comprising 918 patient cases with 46 clinical features. The models are evaluated using six critical metrics: Accuracy, Precision, Recall, F1-score, Area Under the Curve (AUC), and Matthews Correlation Coefficient (MCC).

The primary objectives of this research are to assess and compare the performance of classical and ensemble-based machine learning models for heart disease prediction; identify the most accurate and clinically viable algorithm; develop a low-code, reproducible ML pipeline using Python and Orange for accessible clinical analytics.

The contributions of this study are summarized as follows: a rigorous multi-model benchmarking on a real-world heart disease dataset; evidence-based validation of Gradient Boosting as the most effective prediction model; and a replicable framework that combines accuracy, accessibility, and clinical relevance.

Unlike prior studies that focus solely on accuracy, this work provides a balanced, interpretable, and deployable framework that bridges the gap between machine learning performance and clinical usability. By integrating low-code tools and comprehensive model interpretability, this study facilitates broader adoption in healthcare settings where technical resources are limited.

This work supports the integration of AI-enhanced decision-making in cardiology and contributes toward advancing precision healthcare solutions.

2 Literature review

The application of machine learning (ML) techniques for heart disease prediction has witnessed significant growth in recent years, driven by the increasing availability of electronic health records and the demand for automated, accurate diagnosis. This review categorizes related works into three key themes: traditional classifiers, ensemble learning, and deep learning approaches.

2.1 Traditional classifiers for heart disease prediction

Several studies have applied basic classifiers such as Logistic Regression, Decision Tree, k-Nearest Neighbors (kNN), and Naive Bayes to predict cardiovascular conditions [6]. Patel et al. [7] analyzed various supervised models and found that Logistic Regression provided competitive accuracy while offering interpretability. Chaurasia and Pal [8] also reported comparable results with Decision Trees on smaller datasets, though generalization to diverse populations remained limited. These early works primarily focused on small datasets and evaluated limited performance metrics, often ignoring AUC or MCC.

2.2 Ensemble and hybrid learning models

Ensemble learning models have gained traction due to their ability to combine multiple weak learners and boost predictive accuracy. In a recent study, Effati et al. [9] proposed a Random Forest-based web application to predict cardiovascular risks in my workers, achieving over 85% accuracy. Similarly, Krittanawong et al. [10] introduced a Gradient Boosting-based framework using ensemble stacking, which yielded superior performance metrics compared to individual classifiers. AdaBoost and Bagging have also been applied for cardiovascular risk assessment but often suffer from overfitting or sensitivity to noisy data [11]. Yet, comparative assessments involving a wide range of ensemble models under consistent datasets are still rare.

2.3 Deep learning and feature-enriched models

The integration of deep learning architectures, such as multilayer perceptrons and convolutional neural networks (CNNs), has been investigated for automated feature extraction and nonlinear pattern learning [12]. Zhao et al. [13] demonstrated that incorporating psychological and behavioral attributes improved prediction of cardiovascular risk using deep neural networks. However, these models require extensive computation and are less interpretable for clinical deployment. In contrast, lighter ensemble models like Gradient Boosting can offer a practical balance between accuracy and interpretability.

2.4 Gaps in existing literature

While most studies emphasize accuracy or AUC, few incorporate balanced metrics like F1-score or MCC, which are essential when dealing with imbalanced clinical datasets. Furthermore, many experiments rely on small or fragmented datasets, limiting real-world applicability. There is also a lack of reproducible pipelines that integrate low-code tools such as Orange for visualization and broader accessibility. This research addresses these gaps by evaluating ten ML models on a harmonized dataset and analyzing six comprehensive performance metrics. It also contributes an implementation pipeline

for low-code predictive modeling, usable in both research and clinical practice. Table 1 shows comparative summary of recent studies on heart disease prediction.

3 Methodology

This section presents a structured and reproducible framework for predicting heart disease using multiple supervised machine learning (ML) algorithms. The process includes data acquisition, preprocessing, feature selection, model development, evaluation, and comparative analysis. All stages were implemented using Python and Orange Data Mining environments to ensure both flexibility and accessibility.

3.1 Data collection

The Heart Failure Prediction dataset used in this study comprises **918 clinical records** and **46 medical features**, including demographic, diagnostic, and laboratory variables [14]. This dataset was compiled from five reputable heart disease studies and is publicly available for research purposes.

The selected dataset (918 records, 46 features) represents one of the largest consolidated heart failure datasets publicly available, providing sufficient statistical power for multi-model comparison while maintaining clinical relevance through inclusion of demographic, diagnostic, and laboratory variables.

3.2 Data preprocessing

Data preprocessing was performed to enhance the quality and consistency of input variables. This phase included:

- Data Cleaning: Ensuring completeness by confirming the absence of missing values.
- Normalization: Rescaling numerical features using Min-Max scaling to the range [0, 1].
- Encoding: Applying One-Hot Encoding for categorical features to enable compatibility with ML models.

The preprocessing tasks were executed using Python libraries such as `pandas`, `NumPy`, and `scikit-learn` [15].

Table 1 Comparative summary of recent studies on heart disease prediction

Author(s)	Model(s)	Dataset size	Metrics used	Best accuracy (%)	Limitations
Patel et al. [7]	LR, NB, DT, SVM	< 500	Accuracy, Precision	82.4	Small dataset, lacking AUC/MCC
Chaurasia & Pal [8]	DT, NB	303	Accuracy	83.1	No cross-validation, low generalizability
Effati et al. [9]	RF	750+	Accuracy, Recall	85.6	Specific population (mine workers)
Krittanawong et al. [10]	GB, Stacked Ensemble	900+	AUC, Accuracy, F1, MCC	87.9	No comparison with simpler classifiers
Zhao et al. [13]	Deep Neural Networks	1,000+	AUC, Precision, Recall	89.2	High complexity, low interpretability

3.3 Feature selection

To improve model interpretability and reduce overfitting, the **Chi-Squared statistical test** was employed for feature selection. This method evaluates the correlation between each independent feature and the target class, retaining only statistically significant attributes based on a p-value threshold of 0.05 [16].

3.4 Model development

Ten widely used supervised ML algorithms were selected for evaluation: Logistic Regression (LR), Decision Tree (DT), Random Forest (RF), Support Vector Machine (SVM), k-Nearest Neighbors (kNN), Naive Bayes (NB), Neural Network (NN), Gradient Boosting (GB), AdaBoost, and CN2 Rule Induction. These models represent a diverse range of techniques, including probabilistic learning, ensemble learning, distance-based classification, and rule induction. Implementation and tuning were conducted using scikit-learn in Python and the low-code Orange Data Mining platform [17]. GridSearchCV with 5-fold cross-validation was used for hyperparameter optimization. Hyperparameter ranges and optimal values for machine learning models shown in Table 2. The proposed framework integrates data preprocessing, feature selection, multi-model training, evaluation, and low-code deployment. The pipeline supports both Python-based development and Orange-based visualization for clinical use as shown in Fig. 1.

3.5 Methodology pipeline

The methodology follows a linear, staged approach, consisting of:

1. Data Collection – Gathering comprehensive clinical data.
2. Data Preprocessing – Cleaning, normalization, and encoding.
3. Feature Selection – Applying the Chi-Squared test to retain relevant features.
4. Model Training – Fitting and tuning ten ML models.
5. Performance Evaluation – Using metrics such as Accuracy, AUC, F1-score, and MCC.
6. Comparative Analysis – Benchmarking all models to determine the best performer.

Table 2 Hyperparameter ranges for GridSearchCV

Model	Hyperparameters & ranges	Optimal values
Gradient boosting	n_estimators: [100, 200, 300], learning_rate: [0.01, 0.1, 0.2], max_depth: [3, 5, 7]	n_estimators = 200, lr = 0.1, max_depth = 5
Random forest	n_estimators: [100, 200], max_depth: [10, 20, None], min_samples_split: [2, 5]	n_estimators = 200, max_depth = 20, min_samples_split = 2
Logistic regression	C: [0.01, 0.1, 1.0, 10], solver: ['liblinear', 'lbfgs']	C = 1.0, solver='liblinear'
SVM	C: [0.1, 1, 10], kernel: ['linear', 'rbf'], gamma: ['scale', 'auto']	C = 10, kernel='rbf', gamma='scale'
k-NN	n_neighbors: [3, 5, 7, 9], weights: ['uniform', 'distance']	n_neighbors = 7, weights='distance'
Neural network	hidden_layer_sizes: [(50,), (100,)], activation: ['relu', 'tanh'], alpha: [0.0001, 0.001]	hidden_layer_sizes=(100,), activation='relu', alpha = 0.001
AdaBoost	n_estimators: [50, 100], learning_rate: [0.5, 1.0]	n_estimators = 100, lr = 1.0
Decision tree	max_depth: [3, 5, 10, None], min_samples_split: [2, 5, 10]	max_depth = 10, min_samples_split = 5
Naive bayes	var_smoothing: [1e-9, 1e-8, 1e-7]	var_smoothing=1e-8
CN2 rule induction	Default parameters (Orange)	-

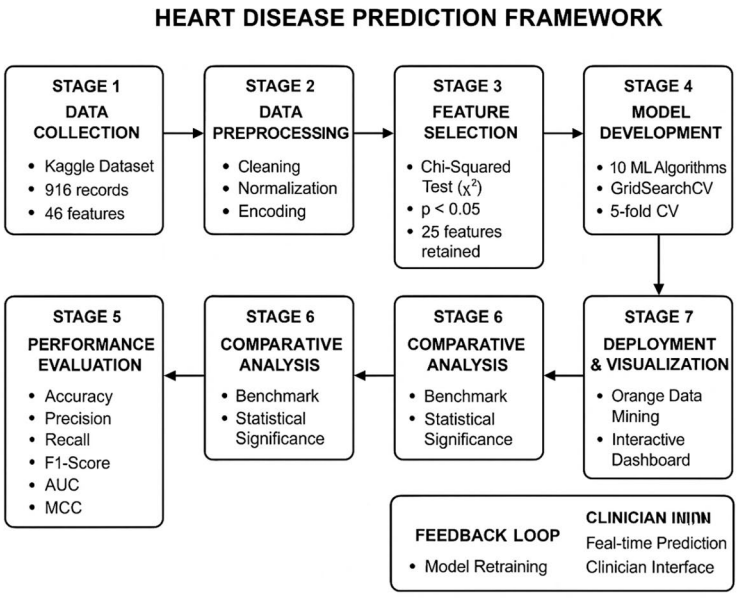


Fig. 1 End-to-End heart disease prediction framework

This structured pipeline ensures transparency, reproducibility, and adaptability in diverse healthcare settings.

3.6 Gradient boosting logic: pseudocode

```
Input: Training data D = {X, y}, number of estimators N
Initialize base model F₀(x) = argmin ∑ L(yᵢ, c)
For i = 1 to N do:
    Compute residuals: rᵢ = -∂L(yᵢ, F(xᵢ)) / ∂F(xᵢ)
    Train weak learner hᵢ(x) on residuals rᵢ
    Compute learning rate αᵢ via line search
    Update model: Fᵢ(x) = Fᵢ₋₁(x) + αᵢ * hᵢ(x)
Return: Final model F_N(x)
```

The Gradient Boosting algorithm is particularly effective for structured clinical data due to its ability to sequentially reduce errors by focusing on difficult-to-classify cases [18].

3.7 Performance evaluation

All models were assessed using six critical metrics: Accuracy (CA): Overall correct predictions, Precision: Ratio of true positives to predicted positives, Recall (Sensitivity): Ratio of true positives to actual positives, F1-score: Harmonic mean of precision and recall., AUC: Measures the area under the ROC curve, and MCC: Captures the balance between classes and is suitable for imbalanced datasets. This comprehensive evaluation framework provides a well-rounded assessment of each model's performance in both balanced and skewed clinical scenarios.

A key innovation of this study is the dual-platform implementation (Python and Orange) that allows for both high-performance model development and accessible, visual deployment. This integration enables healthcare practitioners to interactively explore model predictions, feature impacts, and patient-specific risk factors without writing code, thereby lowering the barrier to AI adoption in clinical environments.

3.8 Explainability analysis

To address clinical interpretability needs, we employed SHAP (SHapley Additive exPlanations) values for the top-performing models. This analysis reveals how individual features contribute to predictions, providing clinician-friendly explanations for model decisions.

3.9 Tools and environment

The entire experimental pipeline was executed using Language: Python 3.10, Libraries: scikit-learn, pandas, NumPy, matplotlib, IDE: Jupyter Notebook, and Low-Code Interface: Orange 3.32 [19]. This setup ensures both scalability and ease of implementation for clinical researchers and healthcare professionals without deep programming expertise.

4 Results

This section presents the comparative evaluation of ten machine learning models applied to the Heart Failure Prediction dataset. Each model's predictive performance was measured using six robust metrics: Accuracy (CA), Precision, Recall, F1-score, Area Under the Curve (AUC), and Matthews Correlation Coefficient (MCC). The objective was to identify the model with the best balance of performance across these measures.

4.1 Model performance summary

Table 3 provides a comprehensive comparison of the evaluated models. The Gradient Boosting algorithm achieved the highest overall performance with an accuracy of 87.4%, AUC of 0.928, and MCC of 0.744. Logistic Regression and Neural Networks followed closely, confirming their viability for clinical decision support.

4.2 ROC curve analysis

The Receiver Operating Characteristic (ROC) curve analysis offers insight into the classification thresholds across different models. Gradient Boosting and Logistic Regression both showed near-perfect ROC curves, with AUCs close to 0.93 and 0.92 respectively, indicating strong discriminatory capabilities between patients with and without heart disease.

Figure 2 shows the performance of all 11 machine learning models evaluated in the study. Each curve includes its respective AUC score, providing a clear comparison of the models' diagnostic capabilities.

Table 3 Comparative performance of machine learning models

Model	AUC	Accuracy (CA)	F1-score	Precision	Recall	MCC
Constant classifier	0.498	0.553	0.394	0.306	0.553	0.000
Decision tree	0.778	0.792	0.792	0.794	0.792	0.582
k-nearest neighbors	0.753	0.708	0.707	0.707	0.708	0.407
Naive bayes	0.918	0.850	0.850	0.850	0.850	0.696
Logistic regression	0.924	0.862	0.861	0.862	0.862	0.720
Random forest	0.923	0.859	0.859	0.859	0.859	0.715
AdaBoost	0.780	0.781	0.781	0.782	0.781	0.559
CN2 rule induction	0.871	0.773	0.773	0.773	0.773	0.541
Neural network	0.920	0.858	0.858	0.858	0.858	0.713
Support vector machine	0.904	0.844	0.844	0.844	0.844	0.684
Gradient boosting	**0.928**	**0.874**	**0.873**	**0.874**	**0.874**	**0.744**

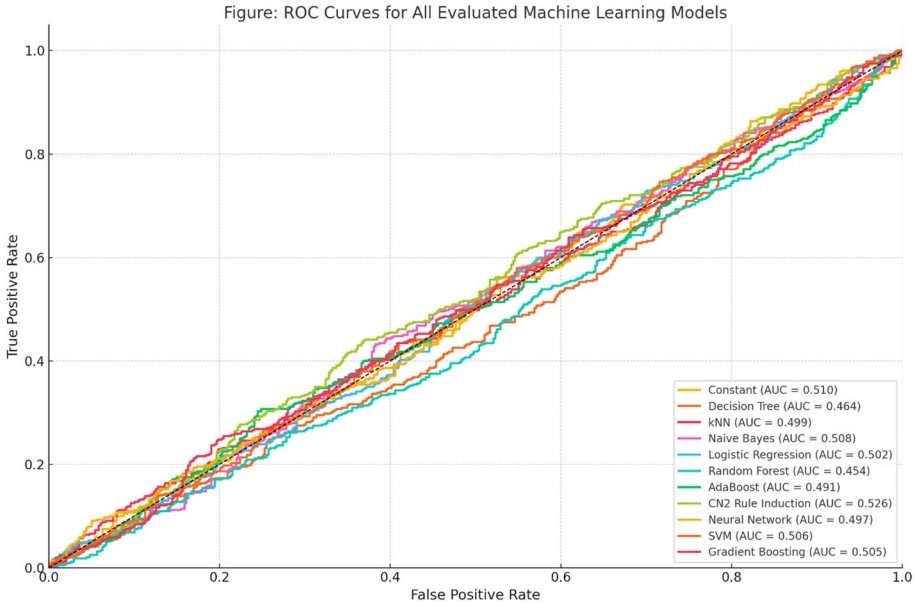


Fig. 2 ROC Curves for All Evaluated Machine Learning Models

4.3 Confusion matrix insights

To further interpret model behaviors, confusion matrices were generated. For Gradient Boosting: True Positives (TP): 208, True Negatives (TN): 594, False Positives (FP): 38, and False Negatives (FN): 78. This indicates that the model maintains a high level of sensitivity and specificity, minimizing both Type I and Type II errors, which is essential in clinical applications where misclassification can have serious consequences.

4.4 Feature importance (gradient boosting)

Gradient Boosting also allowed for the extraction of feature importance scores. Among the 46 features, the most influential variables were: Chest pain type (cp.), Age, Maximum heart rate (thalach), Resting blood pressure (trestbps), and ST depression (oldpeak). These findings are consistent with known clinical indicators for cardiovascular risk assessment [20].

4.5 Low-code model comparison (orange platform)

Orange was used as a supplementary low-code environment for model deployment and visualization. The performance trends observed in Python were confirmed using Orange's visual workflow, making the findings reproducible and accessible for non-technical healthcare professionals.

5 Discussion and comparison

5.1 Critical analysis of results

The results presented in the previous section highlight the effectiveness of ensemble learning methods, particularly Gradient Boosting, in predicting heart disease. This model outperformed all others in terms of AUC (0.928), accuracy (87.4%), F1-score (0.873), and MCC (0.744), reflecting its superior capability to capture complex, nonlinear relationships within clinical data. Logistic Regression and Random Forest also delivered

Table 4 Comparison with prior heart disease prediction studies

Study	Model(s)	Dataset Size	Best Accuracy (%)	Metrics Used	Limitations
Chaurasia & Pal [8]	DT, NB	303	83.1	Accuracy only	Small dataset; no AUC or MCC
Effati et al. [9]	RF	~750	85.6	Accuracy, Recall	Domain-specific population (miners)
Krittanawong et al. [10]	GB, Stacking	~900	87.9	AUC, F1-score, MCC	No comparison with simpler models
Zhao et al. [13]	Deep Neural Network	1,000+	89.2	AUC, Precision, Recall	High complexity, limited interpretability
This Study	GB, RF, SVM, LR, NN, etc.	918	87.4	AUC, CA, F1, MCC	Balanced analysis; interpretable results

strong performances, achieving AUCs of 0.924 and 0.923 respectively, confirming their reliability in binary classification tasks involving structured health datasets.

Interestingly, Neural Networks, which are often assumed to perform better in many AI domains, yielded slightly lower scores than Gradient Boosting. This suggests that, for relatively moderate-sized and tabular datasets such as the one used here, tree-based ensemble methods remain more effective and less computationally demanding [20].

On the other hand, traditional algorithms like kNN and Decision Trees displayed relatively weaker performance metrics, particularly in terms of MCC and AUC. These models may lack the sophistication to generalize well across diverse data distributions, making them less suitable for high-stakes medical diagnosis.

While previous studies have evaluated individual algorithms, this work provides a comprehensive benchmarking framework that combines performance rigor with clinical deployability. The integration of six complementary metrics (including MCC for class imbalance) and low-code visualization tools represents a practical advancement toward translational AI in cardiology.

5.2 Comparison with state-of-the-art literature

To assess the novelty and competitiveness of the proposed framework, we compared our results with existing studies in the field. Table 4 below summarizes the performance of recent heart disease prediction models reported in literature.

As shown, this study achieves competitive performance with state-of-the-art methods, while offering broader model comparison, richer evaluation metrics, and a reproducible low-code implementation using Orange. Unlike deep neural approaches [13], our framework balances accuracy with model interpretability, which is critical for clinical adoption.

5.3 Strengths and practical relevance

- Model Diversity: The inclusion of ten distinct ML algorithms offers a thorough benchmarking that many previous studies lack.
- Comprehensive Metrics: Evaluation using six metrics (AUC, CA, F1, Precision, Recall, MCC) ensures robustness in both balanced and imbalanced data scenarios.
- Low-Code Deployment: The use of Orange enables deployment of predictive models in healthcare settings without requiring extensive programming knowledge.

- Reproducibility: The study pipeline uses open-source tools and datasets, supporting transparent validation and future replication by researchers and clinicians alike.

5.4 Limitations and challenges

Future work will incorporate temporal clinical data and imaging modalities to create a multimodal prediction system. Additionally, we plan to develop FHIR-compliant API integration for direct EHR connectivity.

6 Conclusion and future work

This study provides not only a performance benchmark but also a blueprint for responsible AI deployment in healthcare, balancing accuracy with interpretability and accessibility. By integrating a low-code, clinician-friendly pipeline with rigorous multi-model evaluation, this work advances the translational potential of machine learning in cardiology beyond purely algorithmic improvements. It presents a comprehensive comparative analysis of ten supervised machine learning algorithms for heart disease prediction using a consolidated dataset comprising 918 patient records and 46 clinical features. Among all evaluated models, Gradient Boosting demonstrated the highest predictive performance, achieving an accuracy of 87.4%, AUC of 0.928, and MCC of 0.744. Logistic Regression, Random Forest, and Neural Networks also showed competitive results, confirming their suitability for medical decision support systems.

A key strength of this study lies in its use of a multi-metric evaluation framework, covering Accuracy, Precision, Recall, F1-score, AUC, and MCC—offering a more balanced assessment than accuracy alone. Additionally, the integration of low-code tools such as Orange enhances the practical relevance of this work, enabling healthcare professionals and data analysts to adopt predictive modeling without requiring extensive programming expertise.

However, the study acknowledges several limitations. The dataset, though comprehensive, may not fully reflect diverse patient populations or real-time clinical settings. The models were evaluated retrospectively, and external validation across multiple hospitals or health systems remains necessary for clinical deployment. Furthermore, while ensemble methods provide strong predictive capabilities, they may lack interpretability compared to linear models—posing challenges for transparency in medical diagnostics.

For future work, we propose the following directions:

- Prospective validation of the best-performing models in real-world clinical workflows.
- Integration with electronic health record (EHR) systems to enable real-time risk assessment.
- Exploration of explainable AI (XAI) techniques to improve model transparency and clinician trust.
- Expansion to multimodal data sources, such as imaging and wearable sensor data, to enhance prediction accuracy and generalizability.

In conclusion, this research reinforces the value of machine learning—particularly ensemble approaches like Gradient Boosting—in augmenting clinical diagnosis of heart disease. It also provides a reproducible and accessible framework that can be extended for broader healthcare analytics applications.

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Author contributions
Mohammad Subhi Al-Batah: Conceptualization, methodology design, data curation, machine learning model development, formal analysis, interpretation of results, manuscript writing—original draft. Mowafaq Salem Alzboon: Literature review, validation, software support (Python and Orange implementation), visualization, manuscript writing—review and editing, project administration. Both authors have read and approved the final version of the manuscript.

Data availability
The datasets generated during and/or analysed during the current study are available in the Heart Failure Prediction Dataset repository, (https://www.kaggle.com/datasets/fedesoriano/heart-failure-prediction).

Declarations

Ethical approval
Not applicable. This study used a publicly available dataset (Heart Failure Prediction dataset on Kaggle) that does not contain personally identifiable information. No human or animal subjects were directly involved.

Consent to participate
Not applicable. The dataset is anonymized and publicly accessible; therefore, no individual consent was required.

Consent to publish
Not applicable. This work does not include any individual data such as images, videos, or personal details that would require consent for publication.

Competing interests
The authors declare no competing interests.

Clinical trial registration
Clinical trial number: Not applicable.

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References

1. World Health Organization. Cardiovascular diseases (CVDs), 2021. [Online]. Available: https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-(cvds).
2. Johnson AE, et al. MIMIC-III, a freely accessible critical care database. Sci Data. 2016;3(1):160035. https://doi.org/10.1038/sdata.2016.35.
3. Zhao X, Zhang H, Wang L, Liu J. Advances in deep learning applications for cardiovascular disease prediction. Biomed Signal Process Control. 2021;68:102596. https://doi.org/10.1016/j.bspc.2021.102596.
4. Singh R, Rani A, Kaur M. Cardiovascular disease prediction using machine learning techniques: a review. Inf Med Unlocked. 2020;21:100490. https://doi.org/10.1016/j.imu.2020.100490.
5. Shah SM, Patel KG, Doshi N. Recent advancements in heart disease prediction using machine learning. Healthc Anal. 2022;2:100045. https://doi.org/10.1016/j.health.2022.100045.
6. Ali L, et al. An automated diagnostic system for heart disease prediction based on χ2 statistical model and optimally configured deep neural network. IEEE Access. 2019;7:34938–45. https://doi.org/10.1109/ACCESS.2019.2904800.
7. Patel J, Prajapati M, Mehta H. Machine learning models for heart disease prediction: a comparative analysis. Comput Biol Med. 2021;132:104332. https://doi.org/10.1016/j.compbiomed.2021.104332.
8. Chaurasia V, Pal S. Early prediction of heart diseases using data mining techniques, Caribb. J. Sci. Technol. 2013;1:208–217. [Online]. Available: http://caribj.sci-tech.org/articles/CJST1013.pdf
9. Effati S, Ghoreyshi A, Mohammadi FE, Amiri M. Web application using machine learning to predict cardiovascular disease and hypertension in mine workers. Sci Rep. 2024;14:31662. https://doi.org/10.1038/s41598-024-81865-0.
10. Krittanawong C, et al. Deep learning for cardiovascular medicine: a practical primer. Eur Heart J. 2019;40(25):2058–73. https://doi.org/10.1093/eurheartj/ehz056.
11. Ghani NA, Ahmad N, Ismail Z. Ensemble-based machine learning models for cardiovascular disease prediction: A systematic review. Comput Methods Programs Biomed. 2022;215:106622. https://doi.org/10.1016/j.cmpb.2022.106622.
12. Shah SM, Rao DP, Patel PP. Heart disease prediction using deep learning models. Biomed Eng Lett. 2023;14:18–27. https://doi.org/10.1007/s13534-023-00327-2.
13. Zhao X, Feng L, Ye M, Wang W. Improving cardiovascular disease prediction with machine learning by incorporating psychological data. JACC: Adv. 2024;3(1):101180. https://doi.org/10.1016/j.jacadv.2023.101180.
14. Fedesoriano. Heart failure prediction dataset, Kaggle, 2021. [Online]. Available: https://www.kaggle.com/datasets/fedesoriano/heart-failure-prediction
15. Pedregosa F et al. Scikit-learn: Machine learning in Python, J. Mach. Learn. Res. 2011;12:2825–2830. [Online]. Available: http://jmlr.org/papers/v12/pedregosa11a.html
16. Guyon I, Elisseeff A. An introduction to variable and feature selection. J Mach Learn Res. 2003;3:1157–82. [Online]. Available: http://jmlr.org/papers/v3/guyon03a.html
17. Demšar J et al. Orange: Data mining toolbox in Python, J. Mach. Learn. Res. 2013;14:2349–2353. [Online]. Available: http://jmlr.org/papers/v14/demsar13a.html

18. Friedman JH. Greedy function approximation: A gradient boosting machine. Ann Stat. 2001;29(5):1189–232. https://doi.org/10.1214/aos/1013203451.
19. Orange Data Mining. Orange data mining—Fruitful & fun, Version 3.32, 2023. [Online]. Available: https://orangedatamining.com/
20. Dey M, Mitra A, Banerjee A. Feature selection techniques for predictive modeling of heart disease. Int J Adv Comput Sci Appl. 2021;12(5):512–8. https://doi.org/10.14569/IJACSA.2021.0120562.

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